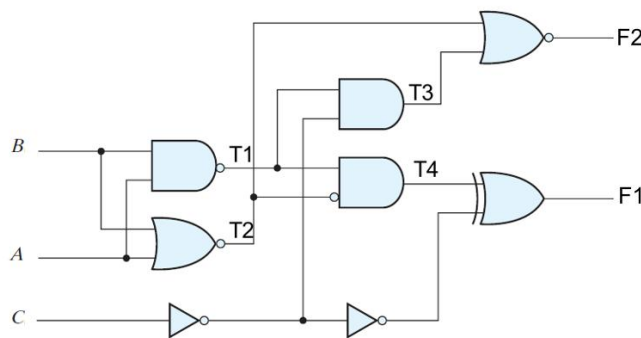


Assignment 2 25F

1. Simplify the following three-variable Boolean functions to simplest **POS** form using K-map method: $F(A,B,C,D) = \Pi(0, 2, 6, 11, 13, 14, 15) + d(1, 3, 9, 10, 12)$.
2. Obtain the simplified Boolean expressions for output F_1 and F_2 in terms of the input variables in the following circuit.
 - a) Derive the Boolean expressions for T_1 through T_4 . Evaluate the outputs F_1 and F_2 as a function of the 3 inputs, you just need to simplify the final output.
 - b) List the truth table of the 3 input variables. Then list the binary values for T_1 through T_4 and outputs F_1 and F_2 in the table.



3. Design a combinational circuit with three inputs, x , y , and z , and three outputs, A , B , and C . When the binary input is 0, 1, or 2, the binary output is two greater than the input. When the binary input is 3, 4, 5, 6, or 7, the binary output is one less than the input. You need to provide:
 - c) List the truth table
 - d) K-map simplification
 - e) Draw the logic diagram
4. A combinational circuit is defined by the equations

$$F1 = AB + A'B'C'$$

$$F2 = A + B + C'$$

$$F3 = A'B + AB'$$
 Design a circuit which implements these three equations using a decoder and OR gates external to the decoder, and draw the logic diagram.
5. (20 points 8+6+6) An 8:1 multiplexer has inputs A , B , and C connected to the selection inputs S_2 , S_1 , and S_0 , respectively. The data inputs I_0 through I_7 are as follows: $I_1 = I_2 = 0$; $I_3 = I_5 = I_7 = 1$; $I_0 = I_4 = D$; and $I_6 = D'$. Determine the Boolean function $F(A, B, C, D)$ that the multiplexer implements. You need to:
 - f) Write down the truth table
 - g) Simplify the Boolean function with K-map
 - h) Implement the following Boolean function with only one 4x1 multiplexer and

external basic gates. (Connect inputs A and B to the selection lines. The input requirements for the four data lines will be a function of variables C and D. These values are obtained by expressing F as a function of C and D for each of the four cases when $AB = 00, 01, 10, \text{ and } 11$. These functions may have to be implemented with external gates.)

6. Construct a 16x1 multiplexer with two 8x1 and one 4x1 multiplexers.
7. Draw the circuit specified by the following HDL description, use at least one standard component.

```
module multiplexer(  
    output      F,  
    input       a,  
    input       b,  
    input       c,  
    input       d,  
    input [1:0] sel  
);  
    assign F = sel[0] ? (sel[1] ? d : b)  
                : (sel[1] ? c : a);  
endmodule
```